COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 | Algorithm Complexity	**SECTION:**	CPE21S2
DATE PERFORMED:	21/10/2026	**DATE SUBMITTED:**	
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ACTIVITY NO. 14.1 **Algorithm Complexity**

1. PROCEDURE OUTPUT

	Best Case	Worst Case	Analysis
Merge Sort	O(n log n)	O(n log n)	If the merge sort were to sort values of n: 100, 1,000, and 10,000; it would have to continuously half the value, as during the sorting as the splitting and merging cost is input-independent which is practically the same for its worst case.
Quick Sort	O(n log n)	$O(n^2)$	The pivot of the quick sort would repeatedly split the array evenly thus the O(n log n) of its quick sort and the worst case would be $O(n^2)$ because of the pivot as it will always be the smallest or largest element.

Table 14-1. Best- and Worst-Case Analysis using Theoretical Tools

	Merge Sort	Quick Sort
Best Case	Merge Sort (already sorted): 9700 ns	Quick Sort: 1100 ns
Worst Case	Shell Sort: 800 ns Merge Sort: 17500 ns	Quick Sort (already sorted): 1500 ns
Analysis	The worst case of the merge sort arrives at 9700 ns while its worst case is 17,500 ns this was due to the splitting and merging of array. In simple terms, it did not have to swap or shift places with the other values leading to the best case of 9700ns.	The already sorted array takes longer due to a wrong/poor choice of the pivot, it could split the array into one empty side and the other side with the n-1. In simple terms, one side could have been left empty and the other one the right side with all the other elements, thus needing it to sort the array all over again.

Table 14-2. Best- and Worst-Case Analysis using Experimental Tools

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Pseudocode Analysis	
Binary Search	**Linear Search**
BinarySearch(A, n, key) low ← 0 high ← n-1 while low ≤ high mid ← low + (high - low) / 2 if A[mid] = key return mid else if A[mid] < key low ← mid + 1 else high ← mid - 1 return -1	LinearSearch(A, n, key) for i ← 0 to n-1 if A[i] = key return i return -1
Analysis	
The binary search grabs the highest and lowest on the array while also getting the middle before it searches by comparing the target to the middle element and if they match, it would be done. Meanwhile if the target is smaller it would the right half and if the target is higher then it would cut off the left side.	The learn search checks every element one at a time, starting from the first index until it finds the element or there are no more elements inside the array.

Time and Space Complexities	
Linear Search	**Binary Search**
Best Case: 0(1) **Worst Case:** O(n)	**Best Case:** 0(1) **Worst Case:** O(log n)

Input Size (N)	Elapsed Time for Binary Search (Worst-Case)	Elapsed Time for Linear Search (Worst-Case)
1000	48.7	3590.1
10,000	68.8	36609.9
100,000	94.6	367094

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Input Size (N)Binary Search             Linear Search
1000          48.7                      3590.1
10000         68.8                      36609.9
100000        94.6                      367094
```

ANALYSIS
The linear search grows by 10x each increase of the N size, while the binary search is roughly around 100x but the time barely changed. The grap between the two widens as N grows. Binary search wins on large inputs, but it needs sorted data, and sorting costs O(n log n).

CONCLUSION	
LOPEZ	To conclude this activity, we returned to the topic of time complexity, so there wasn't much to talk about because it had already been covered. In addition, the process uses the chrono library—more especially, the nanoseconds—to investigate theoretical and experimental tools for time complexity and runtime. This also applies to the supplementary, although there were a few minor adjustments made to the linear and binary search. changes, but the chrono—nanoseconds—remains essentially unchanged. Overall, the task was not overly complicated and only required a few small tweaks, such as adding a library and a function

	for it to function.
REALUBIT	In this hands-on activity, we went back to time complexities thus there were not much for the discussion as it has already been discussed. Apart from that the procedure explores the theoretical and experimental tools for the time complexity and runtime using the chrono library and specifically the nanoseconds. The same goes for the supplementary with some slight changes which were the linear and binary search. changes but essentially still the same with the chrono, nanoseconds. Overall, the activity was not too complex and only needed some minor adjustments where it required only adding a library and a function of that library to run.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Claude AI**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Code Assistance, Explanation of Concepts**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **The AI was explicitly used only for errors that were unfamiliar and the concepts were mainly forgotten concepts similar to the time and space complexity.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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